Fixed-Income and Credit Risk

Midterm Practice Paper

Questions and Detailed Worked Solutions

Main paper: Questions 1–5, 100 marks, suggested time 120 minutes. Attempt all five questions. Show your calculations and justify each derivation. A calculator or software capable of computing a matrix exponential may be used for Question 1(c). Round final monetary values to four decimal places and probabilities to six decimal places, unless otherwise specified.

Additional practice: Questions 6–9 are optional, 20 marks each. They cover further material from the supplied lectures.

Conventions. Time is measured in years; interest rates and bond yields are continuously compounded. All expectations used for valuation are under the risk-neutral (risk-adjusted) measure, denoted by $\mathbb{E}^*$ and $\mathbb{P}^*$. Unless stated otherwise, default terminates future coupons. Φ denotes the standard normal cumulative distribution function. In Questions 4–7, $D(V; V_B)$ denotes a market value of debt; in Questions 3 and 6, D denotes the debt's face value.

The main paper follows the five-question structure of the supplied `midterm2026.pdf`, using new data and some additional subparts. The optional questions extend the same calculation-and-proof style. All questions precede the solutions.

Question	Main topic	Marks
1	Empirical and generator-based rating transitions	20
2	Default probabilities inferred from coupon-bond prices	20
3	Lognormal tax benefits and bankruptcy costs	20
4	First passage, firm valuation and aggregate debt	20
5	Smooth pasting and the endogenous default barrier	20
6	Merton debt, equity and credit spreads	20
7	Optimal debt level after the default-barrier choice	20
8	CDS premium, accrual and protection payments	20
9	Reduced-form pricing and default intensity	20

Important modeling distinction. Question 2 explicitly uses the linear expected-loss approximation of the sample midterm. Question 8 uses exact discrete conditional default probabilities and survival factors. These are different conventions; each is stated in its question.

Part I: Main practice paper

Question 1 (20 marks)

There are four credit classes A, B, C, D, with D the absorbing default state. At time 0 there are 24 firms in each of A, B, C, and none in D. Observe the firms over one year. The only transitions are:

(i) At the end of month 2, two originally B-rated firms move to A, and one other originally B-rated firm moves to C.

(ii) At the end of month 4, three originally A-rated firms move to B.

(iii) At the end of month 8, two originally C-rated firms move to B, and three other originally C-rated firms default.

(iv) At the end of month 11, two other originally C-rated firms default.

All firms mentioned are distinct. Every migrating firm remains in its new state until year-end. All other firms retain their initial rating.

(a) Compute the empirical one-year transition matrix $\widehat{P}_{\text{empirical}}$, using the order A, B, C, D. [5 marks]

(b) Compute the total exposure times N_A, N_B, N_C in firm-years, the transition counts M_{ij}, and the generator estimator $\widehat{\Lambda}$, with off-diagonal entries $\widehat{\lambda}_{ij} = M_{ij}/N_i$. [9 marks]

(c) Compute $\widehat{P}_{\text{generator}}(1) = \exp(\widehat{\Lambda})$. Explain why its (A, D) entry is positive even though no originally A-rated firm defaulted in the sample. [6 marks]

Question 2 (20 marks)

A company has issued two bonds, with maturities 2 and 4 years, respectively. Each bond has face value 100 and pays an annual coupon of 4.2 at the end of each year. Their continuously compounded yields are 4.3% and 4.8%, respectively. The risk-free rate is 2.5% at all maturities, and recovery is 35% of face value, paid immediately upon default.

Default can occur only at times $0.5, 1.5, 2.5, 3.5$. Let P_1 apply to years 1–2 and P_2 to years 3–4. For this question use the *linear expected-loss approximation*: weight the discounted loss at each possible default time by P_1 or P_2 directly, without multiplying by an earlier survival probability. Thus P_i is an approximate annual default probability coefficient, not an exact continuous-time hazard rate.

(a) Compute the corporate and default-free prices of both bonds. [4 marks]

(b) For the 2-year bond, construct a table of remaining default-free value, loss given default, discount factor and discounted loss at each default time. Infer P_1. [7 marks]

(c) Construct the corresponding table for the 4-year bond and infer P_2. Explain why the first two rows must use P_1. [7 marks]

(d) State the survival adjustment needed if P_1, P_2 were interpreted as exact annual conditional default probabilities. [2 marks]

Question 3 (20 marks)

Under the risk-neutral measure, a firm's asset value satisfies

$$dV_t = rV_t\,dt + \sigma V_t\,dW_t, \qquad V_0 > 0,$$

where r, σ are constant and $\sigma > 0$. The firm has one debt payment of face value $D > 0$ due at time $T > 0$; default occurs only at maturity if $V_T < D$.

If the firm survives, it receives a tax benefit κD at time T. If it defaults, a fraction $1 - \alpha$ of its assets is lost as bankruptcy cost, and debtholders receive αV_T. Assume $0 < \kappa < 1$ and $0 < \alpha < 1$. Define

$$A(0) = \mathbb{E}^*[e^{-rT}\kappa D\,\mathbf{1}_{\{V_T \geq D\}}], \qquad C(0) = \mathbb{E}^*[e^{-rT}(1-\alpha)V_T\,\mathbf{1}_{\{V_T < D\}}].$$

The debt and equity payoffs are, respectively,

$$D\mathbf{1}_{\{V_T \geq D\}} + \alpha V_T \mathbf{1}_{\{V_T < D\}}, \qquad (V_T - D + \kappa D)\mathbf{1}_{\{V_T \geq D\}}.$$

(a) Use Itô's lemma to derive the distribution of $\log V_T$ and the risk-neutral probability of default. [5 marks]

(b) Derive closed-form expressions for $A(0)$ and $C(0)$. Show the completion-of-the-square step needed for the truncated expectation in $C(0)$. [8 marks]

(c) Derive debt and equity values $B(0)$ and $S(0)$ and prove $B(0) + S(0) = V_0 + A(0) - C(0)$. [4 marks]

(d) Evaluate $A(0), C(0), B(0), S(0)$ for

$$V_0 = 120, \quad D = 100, \quad r = 0.04, \quad \sigma = 0.25, \quad T = 2, \quad \kappa = 0.20, \quad \alpha = 0.60.$$

[3 marks]

Question 4 (20 marks)

Consider the following first-passage model under the risk-neutral measure:

$$\frac{dV_t}{V_t} = (r-\delta)\,dt + \sigma\,dW_t, \qquad V_0 = V > V_B > 0.$$

Let

$$\tau = \inf\{t \geq 0 : V_t \leq V_B\}, \qquad \xi = \frac{r-\delta}{\sigma^2} - \frac{1}{2},$$

with $r > 0$, $\delta \geq 0$, $\sigma > 0$.

Debt has total face value $P > 0$, continuous coupon rate $\rho > 0$ per unit face value, and maturity density me^{-mT} with $m > 0$. At default, debtholders as a whole receive αV_B, where $0 < \alpha < 1$. An individual bond with unit face value and maturity $T > \tau$ receives $ce^{-r(T-\tau)}$ immediately at default, where

$$c = \frac{m+r}{m}\frac{\alpha V_B}{P}.$$

Assume the chosen parameters and barrier satisfy $0 \leq c \leq 1$. The tax rate is $0 < \kappa < 1$.

(a) Derive, for $s > 0$,

$$\mathbb{E}^*[e^{-s\tau}] = \left(\frac{V_B}{V}\right)^{\sqrt{\xi^2+2s/\sigma^2}+\xi}.$$

Use $e^{-s\tau} = 0$ on $\{\tau = \infty\}$. [4 marks]

(b) Starting from

$$v(V;V_B) = V + \kappa\rho P\,\mathbb{E}^*\left[\int_0^\tau e^{-rt}\,dt\right] - (1-\alpha)V_B\mathbb{E}^*[e^{-r\tau}],$$

derive a closed-form expression for $v(V;V_B)$. [5 marks]

(c) For a unit-face-value bond, let

$$B(V;V_B,T) = \mathbb{E}^*\left[e^{-rT}\mathbf{1}_{\{\tau>T\}} + e^{-r\tau}ce^{-r(T-\tau)}\mathbf{1}_{\{\tau\leq T\}} + \int_0^{\tau\wedge T}\rho e^{-rt}\,dt\right].$$

Evaluate $D(V;V_B) = Pm\int_0^\infty e^{-mT}B(V;V_B,T)\,dT$. Show the principal, coupon and recovery integrals separately. [8 marks]

(d) Verify $v(V_B;V_B) = D(V_B;V_B) = \alpha V_B$, hence $S(V_B;V_B) = 0$, where $S = v - D$. [3 marks]

Question 5 (20 marks)

Use the setting and notation of Question 4. To simplify the expressions, put

$$q = \sqrt{\xi^2 + \frac{2r}{\sigma^2}} + \xi, \qquad h = \sqrt{\xi^2 + \frac{2(r+m)}{\sigma^2}} + \xi.$$

The equity value can be written as

$$S(V;V_B) = V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\frac{V_B}{V}\right)^q - \frac{\rho+m}{r+m}P + \left[\frac{\rho+m}{r+m}P - \alpha V_B\right]\left(\frac{V_B}{V}\right)^h.$$

Assume $r, m, \rho, \sigma > 0$, $0 < \alpha, \kappa < 1$, and $\kappa\rho < r$.

(a) Differentiate $S(V;V_B)$ with respect to V, holding V_B fixed, and evaluate the derivative at $V = V_B$. [5 marks]

(b) Show that the smooth-pasting equation

$$\left.\frac{\partial S(V;V_B^*)}{\partial V}\right|_{V=V_B^*} = 0$$

has the solution $V_B^* = \epsilon P$, where

$$\epsilon = \frac{\dfrac{\rho+m}{r+m}h - \dfrac{\kappa\rho}{r}q}{1 + (1-\alpha)q + \alpha h}.$$

[5 marks]

(c) Prove, under the stated assumptions,

$$0 < \epsilon < \frac{\rho+m}{\alpha(r+m)}.$$

Justify the positivity of the numerator explicitly. [5 marks]

(d) Use

$$r = 0.05, \quad \delta = 0.02, \quad \sigma = 0.20, \quad \rho = 0.06,$$
$$m = 0.20, \quad \kappa = 0.25, \quad \alpha = 0.60, \quad P = 80, \quad V = 150$$

to compute $\epsilon, V_B^*, c, v, D, S$. Verify $V_B^* < V$ and $c < 1$. [5 marks]

Scope of the proof. Parts (a)–(c) establish the smooth-pasting candidate and its bound. A proof that this candidate solves the full limited-liability optimization problem is not required.

Part II: Additional practice

Question 6 (20 marks)

In the Merton model, $dV_t = rV_t\,dt + \sigma V_t\,dW_t$ under the risk-neutral measure. One zero-coupon debt has face value D and maturity T. There are no taxes or bankruptcy costs. Equity receives $(V_T - D)^+$ and debtholders receive $\min(V_T, D)$.

(a) Derive formulas for $S(0)$ and $B(0)$ and show that their sum is V_0. [8 marks]

(b) Derive the risk-neutral default probability, the continuously compounded risky yield y, and the credit spread $y - r$. Explain why $\Phi(-d_1)$ is not the default probability. [5 marks]

(c) Evaluate the preceding quantities for $V_0 = 120, D = 100, r = 0.04, \sigma = 0.25, T = 2$. [3 marks]

(d) If the observed equity return volatility is σ_S, derive the second equation used together with the equity price to infer V_0, σ. [4 marks]

Question 7 (20 marks)

Suppose the first-stage smooth-pasting barrier in Question 5 has been adopted: $V_B^*(P) = \epsilon P$, with $\epsilon > 0$ independent of P. For fixed current asset value V, the second-stage objective is

$$v(V; \epsilon P) = V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r} + (1-\alpha)\epsilon\right] P \left(\frac{\epsilon P}{V}\right)^q, \qquad 0 < P < \frac{V}{\epsilon}.$$

Here $q > 0$, $0 < \alpha, \kappa < 1$, $r, \rho > 0$.

(a) Prove that this objective is strictly concave in P. [6 marks]

(b) Derive the unique maximizing debt level and prove that it lies in $(0, V/\epsilon)$. [8 marks]

(c) For the parameters of Question 5 and $V = 150$, evaluate P^*, $V_B^*(P^*)$, and the maximized firm value. Explain why this derivative with respect to P differs from the smooth-pasting derivative. [6 marks]

Question 8 (20 marks)

Consider a new 4-year CDS with notional 10 million. Premiums are paid annually in arrears at times $1, 2, 3, 4$, provided no default has occurred. The annual premium rate is s. The risk-free rate is 4%, recovery is 35% of notional, and default occurs only halfway through each year.

Conditional on survival to the beginning of each year, the probability of default during that year is exactly $p = 0.025$. On default at time $j - 0.5$, the buyer pays the half-year accrued premium $0.5s$ per unit notional. The seller immediately pays $1 - 0.35$ per unit notional.

(a) Compute the year-end survival probabilities and unconditional mid-year default probabilities. [4 marks]

(b) Construct a discounted cash-flow table for regular premiums, accrued premiums and protection payments, per unit notional. [8 marks]

(c) Find the break-even spread in percent and basis points and the annual premium amount on the stated notional. [6 marks]

(d) Explain the direction of the spread error if accrued premium is omitted. [2 marks]

Question 9 (20 marks)

In the Duffie–Singleton model, recovery is a fraction R of the market value immediately before default. For this question use R for the recovery fraction, reserving δ for asset payout rates in the structural models. Let $r = 0.03$, $R = 0.40$, and let the deterministic risk-neutral default intensity be

$$\lambda(t) = \begin{cases} 0.02, & 0 \le t \le 2, \\ 0.05, & 2 < t \le 5. \end{cases}$$

For a unit-face-value zero-coupon bond, use

$$B(0,T) = \exp\left(- \int_0^T [r + (1-R)\lambda(u)]\, du \right).$$

(a) Compute survival and cumulative default probabilities at years 2 and 5. [5 marks]

(b) Compute the 2-year and 5-year bond prices, yields and credit spreads. [6 marks]

(c) Suppose these two yields and R are known but both piecewise intensities are unknown. Derive the calibration equations and recover the intensities. [5 marks]

(d) If $Q(T) = \mathbb{P}^*(\tau > T)$, express the hazard rate using $Q'(T)$. Explain the role of R in default probability versus bond valuation. [4 marks]

Formula sheet

The following results may be used in the practice paper.

Itô's lemma

If $dX_t = \mu_t\,dt + \sigma_t\,dW_t$ and f is sufficiently smooth, then

$$df(t, X_t) = \left(f_t + \mu_t f_x + \frac{1}{2}\sigma_t^2 f_{xx}\right) dt + \sigma_t f_x\,dW_t.$$

Also, for square-integrable adapted ζ,

$$\mathbb{E}\left[\left(\int_s^t \zeta_u\,dW_u\right)^2\right] = \mathbb{E}\left[\int_s^t \zeta_u^2\,du\right].$$

Normal distribution

$$\phi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}, \qquad \Phi(z) = \int_{-\infty}^{z} \phi(u)\,du, \qquad \Phi(-z) = 1 - \Phi(z).$$

For $Z \sim N(0,1)$, $\mathbb{E}[e^{aZ}] = e^{a^2/2}$.

First-passage transform

For $X_t = \mu t + \sigma W_t$, $\sigma > 0$, $b \neq 0$, and $\tau_b = \inf\{t \geq 0 : X_t = b\}$,

$$\mathbb{E}[e^{-s\tau_b}] = \begin{cases} \exp\left[-\dfrac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} - \mu\right)\right], & b > 0, \\ \exp\left[\dfrac{b}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} + \mu\right)\right], & b < 0, \end{cases} \qquad s > 0.$$

Set $e^{-s\tau_b} = 0$ when $\tau_b = \infty$.

Homogeneous Markov chain

For a generator Λ, the transition matrix over t years is

$$P(t) = e^{t\Lambda} = \sum_{n=0}^{\infty} \frac{t^n \Lambda^n}{n!}.$$

This is a matrix exponential, not entry-by-entry exponentiation.

Part III: Detailed worked solutions

Solution to Question 1

(a) Empirical transition matrix. The empirical estimator compares each firm's initial rating with its year-end rating; it ignores transition times. The initial A cohort has 21 firms remaining in A and 3 ending in B. The initial B cohort has 2 ending in A, 21 in B, and 1 in C. The initial C cohort has 2 ending in B, 17 in C, and 5 in D. Therefore

$$\boxed{\widehat{P}_{\text{empirical}} = \begin{pmatrix} 21/24 & 3/24 & 0 & 0 \\ 2/24 & 21/24 & 1/24 & 0 \\ 0 & 2/24 & 17/24 & 5/24 \\ 0 & 0 & 0 & 1 \end{pmatrix}.}$$

The last row is imposed by the assumption that default is absorbing. It cannot be estimated as a cohort frequency because there were no initially defaulted firms.

(b) Exposure times and generator. Exposure includes time spent in a rating by both initial members and incoming firms. Months must be divided by 12 to obtain firm-years:

$$\begin{aligned} N_A &= 21 + 3\frac{4}{12} + 2\frac{10}{12} = \frac{71}{3} = 23.666667, \\ N_B &= 21 + 3\frac{2}{12} + 3\frac{8}{12} + 2\frac{4}{12} = \frac{145}{6} = 24.166667, \\ N_C &= 17 + 5\frac{8}{12} + 2\frac{11}{12} + 1\frac{10}{12} = 23. \end{aligned}$$

For example, the two firms arriving in A at month 2 contribute $2(12-2)/12$ firm-years to N_A, while the three firms leaving A at month 4 contribute $3(4/12)$.

The nonzero transition counts are

$$M_{AB} = 3, \quad M_{BA} = 2, \quad M_{BC} = 1, \quad M_{CB} = 2, \quad M_{CD} = 5.$$

All other off-diagonal counts are zero. Dividing by the exposure of the *origin* state and setting each diagonal to the negative sum of its off-diagonal row gives, in units of year^{-1},

$$\boxed{\widehat{\Lambda} = \begin{pmatrix} -9/71 & 9/71 & 0 & 0 \\ 12/145 & -18/145 & 6/145 & 0 \\ 0 & 2/23 & -7/23 & 5/23 \\ 0 & 0 & 0 & 0 \end{pmatrix}.}$$

Each row sums to zero and all off-diagonal entries are nonnegative.

Solution to Question 1 (continued)

(c) Generator-based transition probabilities. With the annual time unit,

$$\widehat{P}_{\text{generator}}(1) = e^{\widehat{\Lambda}} = I + \widehat{\Lambda} + \frac{\widehat{\Lambda}^2}{2!} + \cdots .$$

Numerically,

$$\boxed{\widehat{P}_{\text{generator}}(1) \approx \begin{pmatrix} 0.885575 & 0.112075 & 0.002184 & 0.000166 \\ 0.073171 & 0.889392 & 0.033526 & 0.003911 \\ 0.002996 & 0.070453 & 0.739016 & 0.187534 \\ 0 & 0 & 0 & 1 \end{pmatrix}.}$$

The unrounded row sums are 1; small discrepancies in the printed row sums are due to rounding.

Although $\widehat{\lambda}_{AD} = 0$, the fitted homogeneous continuous-time chain allows paths

$$A \longrightarrow B \longrightarrow C \longrightarrow D$$

within one year. Every rate on this path is strictly positive, so the probability of such a path is positive. This explains the positive (A, D) entry. The generator estimator uses the timing information to infer transition rates and permits multiple transitions in a future year.

Useful extension. Under the same homogeneous model, the two-year transition matrix is

$$P(2) = e^{2\widehat{\Lambda}} = \left(e^{\widehat{\Lambda}}\right)^2,$$

which is approximately

$$\begin{pmatrix} 0.792450 & 0.199084 & 0.007306 & 0.001160 \\ 0.129977 & 0.801580 & 0.054754 & 0.013689 \\ 0.010023 & 0.115062 & 0.548514 & 0.326401 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Thus the fitted two-year default probability for a firm currently rated C is 0.326401.

Common errors. Dividing a count by the initial population estimates a cohort probability, not a continuous-time transition rate. Ignoring time spent in a rating by incoming firms gives the wrong generator. Finally, $e^{\widehat{\Lambda}}$ must be computed as a matrix exponential.

Solution to Question 2

(a) Prices from yields. For an integer maturity T, coupon 4.2 and face value 100, the price at continuously compounded yield y is

$$B_y(T) = 4.2\sum_{j=1}^{T} e^{-yj} + 100e^{-yT}.$$

The final coupon is already included in the sum. Hence

T	Default-free price ($y = 0.025$)	Corporate price	Difference
2	103.2144	99.6365	3.5779
4	106.2720	97.4524	8.8196

For instance, $B_{0.043}(2) = 4.2e^{-0.043} + 104.2e^{-0.086}$.

(b) The 2-year bond. At default time t, the value of all remaining promised payments, discounted back to t at the risk-free rate, is

$$F_T(t) = 4.2\sum_{\substack{1\le j\le T\\ j>t}} e^{-0.025(j-t)} + 100e^{-0.025(T-t)}.$$

The immediate recovery is 35, so the loss at t is $F_T(t) - 35$, with time-zero discounted loss $L_T(t) = e^{-0.025t}[F_T(t) - 35]$.

t	$F_2(t)$	$F_2(t) - 35$	$e^{-0.025t}$	$L_2(t)$
0.5	104.5127	69.5127	0.987578	68.6492
1.5	102.9056	67.9056	0.963194	65.4063

For example,

$$F_2(0.5) = 4.2e^{-0.025(0.5)} + 104.2e^{-0.025(1.5)} = 104.5127.$$

Under the stipulated approximation, the price difference equals $P_1[L_2(0.5) + L_2(1.5)]$. Therefore

$$103.2144077 - 99.6365467 = 134.0554861P_1, \qquad \boxed{P_1 = 0.0266894 = 2.66894\%.}$$

Use unrounded numbers when solving for the coefficients.

Solution to Question 2 (continued)

(c) The 4-year bond. The same remaining-payment calculation gives

t	Coefficient	$F_4(t)$	Loss	Discount factor	$L_4(t)$
0.5	P_1	107.6088	72.6088	0.987578	71.7068
1.5	P_1	106.0801	71.0801	0.963194	68.4639
2.5	P_2	104.5127	69.5127	0.939413	65.3011
3.5	P_2	102.9056	67.9056	0.916219	62.2164

The coefficient concerns the calendar year of default, not the maturity of the bond. Thus the first two rows still use P_1. The remaining cash flows are those of the 4-year bond, so their loss coefficients differ from those of the 2-year bond.

The calibration equation is

$$\begin{aligned} 106.2720468 - 97.4524150 &= (71.7068238 + 68.4639406)P_1 \\ &\quad + (65.3011244 + 62.2163985)P_2 \\ &= 140.1707644P_1 + 127.5175229P_2. \end{aligned}$$

Substitution of the unrounded P_1 gives

$$\boxed{P_2 = \frac{8.8196318 - 140.1707644(0.0266894033)}{127.5175229} = 0.0398264 = 3.98264\%.}$$

The fitted later-year coefficient is larger than the early-year coefficient.

(d) Exact conditional-probability convention. If P_1, P_2 were exact annual conditional default probabilities, the unconditional probabilities at the four default times would be

$$P_1, \quad (1-P_1)P_1, \quad (1-P_1)^2P_2, \quad (1-P_1)^2(1-P_2)P_2.$$

These would replace the direct coefficients in the expected-loss equations. One would then solve the adjusted equations again; the values found above are not the exact conditional-probability calibration.

For a continuous-time hazard λ constant over a year, the corresponding one-year conditional default probability is $1 - e^{-\lambda}$, not λ. This is another reason to distinguish a probability from an intensity.

Solution to Question 3

(a) Lognormal distribution and default probability. Apply Itô's lemma to $f(v) = \log v$:

$$d\log V_t = \frac{dV_t}{V_t} - \frac{1}{2V_t^2}(dV_t)^2 = \left(r - \frac{1}{2}\sigma^2\right)dt + \sigma\, dW_t.$$

Integrating from 0 to T,

$$\log V_T = \log V_0 + \left(r - \frac{1}{2}\sigma^2\right)T + \sigma W_T.$$

Since $W_T \sim N(0,T)$,

$$\log V_T \sim N\left(\log V_0 + \left(r - \frac{1}{2}\sigma^2\right)T,\ \sigma^2 T\right).$$

Define

$$d_1 = \frac{\log(V_0/D) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

Standardization gives

$$\boxed{\mathbb{P}^*(V_T < D) = \Phi(-d_2), \qquad \mathbb{P}^*(V_T \geq D) = \Phi(d_2).}$$

The equality event has probability zero because the distribution is continuous.

(b) Tax benefits and bankruptcy costs. The tax benefit is a fixed cash payment conditional on survival:

$$\boxed{A(0) = \kappa D e^{-rT}\Phi(d_2).}$$

To evaluate the asset-weighted default payoff, put $a = \sigma\sqrt{T}$ and write $V_T = V_0 e^{(r-\sigma^2/2)T + aZ}$, with $Z \sim N(0,1)$. The default event is $Z < -d_2$. Therefore

$$\mathbb{E}^*[V_T \mathbf{1}_{\{V_T < D\}}] = V_0 e^{(r-\sigma^2/2)T} \int_{-\infty}^{-d_2} e^{az}\phi(z)\, dz.$$

Complete the square in the exponent:

$$az - \frac{z^2}{2} = -\frac{(z-a)^2}{2} + \frac{a^2}{2}, \qquad e^{az}\phi(z) = e^{a^2/2}\phi(z-a).$$

Using $u = z - a$, the upper integration limit becomes $-d_2 - a = -d_1$. Consequently

$$\mathbb{E}^*[V_T \mathbf{1}_{\{V_T < D\}}] = V_0 e^{rT}\Phi(-d_1), \qquad \boxed{C(0) = (1-\alpha)V_0\Phi(-d_1).}$$

The integral in the standardized variable starts at $-\infty$; this limit is essential when changing variables from a positive lognormal asset value.

Solution to Question 3 (continued)

(c) Debt, equity and total firm value. The discounted asset expectation over the survival region is

$$e^{-rT}\mathbb{E}^*[V_T\mathbf{1}_{\{V_T\geq D\}}] = V_0 - V_0\Phi(-d_1) = V_0\Phi(d_1).$$

Thus the debt value is

$$\boxed{B(0) = De^{-rT}\Phi(d_2) + \alpha V_0\Phi(-d_1),}$$

and the equity value is

$$\boxed{S(0) = V_0\Phi(d_1) - (1-\kappa)De^{-rT}\Phi(d_2).}$$

Adding them and using $\Phi(d_1) + \Phi(-d_1) = 1$,

$$\begin{aligned} B(0) + S(0) &= V_0\Phi(d_1) + \alpha V_0\Phi(-d_1) + \kappa De^{-rT}\Phi(d_2) \\ &= V_0 - (1-\alpha)V_0\Phi(-d_1) + \kappa De^{-rT}\Phi(d_2) \\ &= \boxed{V_0 + A(0) - C(0)}. \end{aligned}$$

This identity also follows directly by adding the terminal debt and equity payoffs in the default and survival regions.

(d) Numerical evaluation. Here $\sigma\sqrt{T} = 0.3535534$ and

$$d_1 = 0.9187341, \qquad d_2 = 0.5651807.$$

Substituting into the closed-form expressions gives

Quantity	Value
Tax benefit $A(0)$	13.1826
Bankruptcy cost $C(0)$	8.5976
Debt $B(0)$	78.8092
Equity $S(0)$	45.7757
Total firm value $B(0) + S(0)$	124.5849

The independent check is $120 + 13.1825551 - 8.5976320 = 124.5849231$.

Common errors. The default probability uses d_2, whereas the truncated asset expectation uses d_1. Do not add an extra discount factor to the recovery term: $e^{-rT}\mathbb{E}^*[\alpha V_T\mathbf{1}_{\{V_T<D\}}]$ already discounts it once.

Solution to Question 4

(a) First-passage transform. The asset process has the solution

$$V_t = V \exp\left[\left(r - \delta - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right].$$

Hence τ is the first time the process $X_t = \mu t + \sigma W_t$ reaches $b = \log(V_B/V) < 0$, where $\mu = r - \delta - \sigma^2/2 = \sigma^2\xi$. The downward first-passage formula yields

$$\begin{aligned}\mathbb{E}^*[e^{-s\tau}] &= \exp\left[\frac{\log(V_B/V)}{\sigma^2}\left(\sqrt{\mu^2 + 2\sigma^2 s} + \mu\right)\right] \\ &= \exp\left[\log(V_B/V)\left(\sqrt{\xi^2 + \frac{2s}{\sigma^2}} + \xi\right)\right] \\ &= \boxed{\left(\frac{V_B}{V}\right)^{\sqrt{\xi^2+2s/\sigma^2}+\xi}}.\end{aligned}$$

For $s > 0$, the exponent is positive, and the transform lies strictly between 0 and 1 when $V > V_B$.

(b) Total firm value. Pathwise, with the same convention when $\tau = \infty$,

$$\int_0^\tau e^{-rt}\,dt = \frac{1 - e^{-r\tau}}{r}.$$

With $q = \sqrt{\xi^2 + 2r/\sigma^2} + \xi$, substitution gives

$$\begin{aligned}v(V; V_B) &= V + \frac{\kappa\rho P}{r}(1 - \mathbb{E}^*[e^{-r\tau}]) - (1-\alpha)V_B\mathbb{E}^*[e^{-r\tau}] \\ &= \boxed{V + \frac{\kappa\rho}{r}P - \left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\frac{V_B}{V}\right)^q}.\end{aligned}$$

The first correction is the value of tax deductions earned until default; the second is the discounted bankruptcy loss.

(c) Aggregate debt: interchange of integrals. All three payoff components are nonnegative, so Tonelli's theorem permits interchanging expectations and maturity integrals. The following calculation first fixes a realized τ, evaluates the integrals in T, and then takes the expectation.

Solution to Question 4 (continued)

Principal component. For a fixed τ,

$$m\int_0^\infty e^{-mT}e^{-rT}\mathbf{1}_{\{T<\tau\}}\,dT = m\int_0^\tau e^{-(m+r)T}\,dT = \frac{m}{m+r}(1-e^{-(m+r)\tau}).$$

Coupon component. Interchanging the maturity integral and the coupon-time integral gives

$$\begin{aligned} m\int_0^\infty e^{-mT}\left(\int_0^{\tau\wedge T}\rho e^{-rt}\,dt\right)dT &= \rho\int_0^\tau e^{-rt}\left(m\int_t^\infty e^{-mT}\,dT\right)dt \\ &= \rho\int_0^\tau e^{-(m+r)t}\,dt \\ &= \frac{\rho}{m+r}(1-e^{-(m+r)\tau}). \end{aligned}$$

Recovery component. Since $e^{-r\tau}ce^{-r(T-\tau)} = ce^{-rT}$,

$$m\int_0^\infty e^{-mT}ce^{-rT}\mathbf{1}_{\{T\geq\tau\}}\,dT = \frac{mc}{m+r}e^{-(m+r)\tau}.$$

Adding the components, taking expectations, and multiplying by P gives

$$D(V;V_B) = P\frac{\rho+m}{r+m}(1-\mathbb{E}^*[e^{-(r+m)\tau}]) + P\frac{mc}{r+m}\mathbb{E}^*[e^{-(r+m)\tau}].$$

Use $Pmc/(r+m) = \alpha V_B$ and $h = \sqrt{\xi^2+2(r+m)/\sigma^2}+\xi$ to obtain the compact result

$$\boxed{D(V;V_B) = \frac{\rho+m}{r+m}P - \left[\frac{\rho+m}{r+m}P - \alpha V_B\right]\left(\frac{V_B}{V}\right)^h.}$$

It is algebraically equivalent to the expanded debt formula in the supplied notes.

(d) Boundary values. At $V = V_B$, both powers equal 1:

$$v(V_B;V_B) = V_B - (1-\alpha)V_B = \alpha V_B, \qquad D(V_B;V_B) = \alpha V_B.$$

Thus $S(V_B;V_B) = 0$. Economically, default is immediate at the barrier, equity gets nothing, and debtholders receive the remaining assets.

Solution to Question 5

(a) Differentiation at fixed barrier. For any constant exponent a,

$$\frac{\partial}{\partial V}\left(\frac{V_B}{V}\right)^a = -\frac{a}{V}\left(\frac{V_B}{V}\right)^a.$$

Here V_B is held fixed. Therefore

$$\begin{aligned} S_V(V;V_B) = 1 &+ \frac{q}{V}\left[\frac{\kappa\rho}{r}P + (1-\alpha)V_B\right]\left(\frac{V_B}{V}\right)^q \\ &- \frac{h}{V}\left[\frac{\rho+m}{r+m}P - \alpha V_B\right]\left(\frac{V_B}{V}\right)^h. \end{aligned}$$

At the boundary this becomes

$$\boxed{S_V(V_B;V_B) = 1 + (1-\alpha)q + \alpha h + \frac{P}{V_B}\left(\frac{\kappa\rho}{r}q - \frac{\rho+m}{r+m}h\right).}$$

(b) Smooth-pasting equation. Set the boundary derivative to zero and multiply by V_B:

$$[1 + (1-\alpha)q + \alpha h]V_B = P\left[\frac{\rho+m}{r+m}h - \frac{\kappa\rho}{r}q\right].$$

Because the coefficient on V_B is strictly positive, the candidate is unique and is

$$\boxed{V_B^* = \epsilon P, \qquad \epsilon = \frac{\dfrac{\rho+m}{r+m}h - \dfrac{\kappa\rho}{r}q}{1 + (1-\alpha)q + \alpha h}.}$$

The operation is a derivative with respect to the current state V followed by evaluation at the barrier. It is not a derivative with respect to V_B.

(c) Positivity and upper bound. Since $m > 0$, $h > q > 0$. To compare the coefficients, write

$$\frac{\rho+m}{r+m} = \frac{r}{r+m}\frac{\rho}{r} + \frac{m}{r+m}.$$

This is a weighted average of ρ/r and 1. If $\rho/r \le 1$, then $(\rho+m)/(r+m) > \rho/r > \kappa\rho/r$. If $\rho/r > 1$, then $(\rho+m)/(r+m) > 1 > \kappa\rho/r$, using $\kappa\rho < r$. Thus in both cases

$$\frac{\rho+m}{r+m} > \frac{\kappa\rho}{r}.$$

Combined with $h > q > 0$, this proves the numerator is strictly positive. The denominator is positive as well, hence $\epsilon > 0$.

Solution to Question 5 (continued)

For the upper bound, dropping the strictly positive subtractive term in the numerator and using $1 + (1-\alpha)q + \alpha h > \alpha h$ gives

$$\epsilon < \frac{\dfrac{\rho+m}{r+m}h}{1+(1-\alpha)q+\alpha h} < \frac{\dfrac{\rho+m}{r+m}h}{\alpha h} = \boxed{\frac{\rho+m}{\alpha(r+m)}}.$$

This bound alone does *not* establish the recovery restriction $c \le 1$, which requires $\epsilon \le m/[\alpha(m+r)]$. For the numerical example we check the recovery restriction directly.

(d) Numerical values. The parameters give

$$\xi = \frac{0.05-0.02}{0.20^2} - \frac{1}{2} = 0.25, \quad q = 1.8507811, \quad h = 3.7943617,$$

and

$$\frac{\rho+m}{r+m} = 1.04, \qquad \frac{\kappa\rho}{r} = 0.30.$$

Hence

$$\epsilon = \frac{1.04(3.7943617) - 0.30(1.8507811)}{1 + 0.40(1.8507811) + 0.60(3.7943617)} = 0.8441527.$$

Consequently

$$V_B^* = 80\epsilon = 67.5322165, \qquad c = \frac{0.25}{0.20}\frac{0.60(67.5322165)}{80} = 0.6331145.$$

Both restrictions hold: $67.5322165 < 150$ and $0 < c < 1$.

Using the formulas from Question 4,

$$v = 150 + 0.30(80) - [0.30(80) + 0.40(67.5322165)]\left(\frac{67.5322165}{150}\right)^{1.8507811},$$

$$D = 1.04(80) - [1.04(80) - 0.60(67.5322165)]\left(\frac{67.5322165}{150}\right)^{3.7943617}.$$

The results, using unrounded intermediate values, are

$$\boxed{v = 162.3524, \qquad D = 81.1338, \qquad S = v - D = 81.2187.}$$

The upper bound is also satisfied: $\epsilon = 0.844153 < 1.04/0.60 = 1.733333$.

Interpretation. The proportional barrier depends on debt face value, maturity profile, volatility, recovery and tax benefits. Smooth pasting characterizes the candidate; it should not be mistaken for a complete proof of the constrained first-stage optimization.

Solution to Question 6

(a) Debt and equity. Using the notation and truncated expectations established in Question 3,

$$\mathbb{E}^*[e^{-rT}V_T\mathbf{1}_{\{V_T\geq D\}}] = V_0\Phi(d_1), \qquad \mathbb{P}^*(V_T \geq D) = \Phi(d_2).$$

Since equity is $(V_T - D)\mathbf{1}_{\{V_T\geq D\}}$,

$$\boxed{S(0) = V_0\Phi(d_1) - De^{-rT}\Phi(d_2).}$$

Debtholders receive D upon survival and the whole asset value on default, so

$$\boxed{B(0) = De^{-rT}\Phi(d_2) + V_0\Phi(-d_1).}$$

Their sum is $V_0[\Phi(d_1) + \Phi(-d_1)] = V_0$. This is the capital-structure irrelevance identity in this idealized model.

(b) Default probability and spread. Default is the event $V_T < D$, hence its probability is $\Phi(-d_2)$. The debt's yield solves $B(0) = De^{-yT}$:

$$\boxed{y = \frac{1}{T}\log\frac{D}{B(0)}, \qquad y - r = \frac{1}{T}\log\frac{De^{-rT}}{B(0)}.}$$

The factor $\Phi(-d_1)$ arises from an *asset-weighted* expectation, not an indicator expectation. Therefore it is not the default probability. Since $\min(V_T, D) \leq D$ pathwise, $B(0) \leq De^{-rT}$, so the spread is nonnegative.

(c) Numerical evaluation. With $d_1 = 0.9187341$ and $d_2 = 0.5651807$,

$$B(0) \approx 87.4069, \qquad S(0) \approx 32.5931, \qquad \mathbb{P}^*(V_T < D) \approx 0.285975.$$

The yield and spread are approximately

$$\boxed{y = 6.7298\%, \qquad y - r = 2.7298\% = 272.98 \text{ basis points.}}$$

(d) Calibration using equity volatility. Regard the equity value as $S(t, V_t)$. By Itô's lemma, its diffusion term is $S_V\sigma V_t\, dW_t$. Dividing by $S(t)$ identifies the equity return volatility:

$$\sigma_S = \frac{\sigma V_0}{S(0)}S_V(0, V_0).$$

The Black–Scholes call delta is $S_V = \Phi(d_1)$. Thus the two calibration equations for the unknown asset value and volatility are

$$\boxed{\begin{aligned} S_{\text{obs}} &= V_0\Phi(d_1) - De^{-rT}\Phi(d_2), \\ \sigma_{S,\text{obs}}S_{\text{obs}} &= \sigma V_0\Phi(d_1). \end{aligned}}$$

The quantities d_1, d_2 depend on the unknown V_0, σ, so the equations are generally solved numerically.

Solution to Question 7

(a) Strict concavity. For clarity put $k = \kappa\rho/r > 0$ and $H = [k + (1-\alpha)\epsilon](\epsilon/V)^q > 0$. Then

$$v(V; \epsilon P) = V + kP - HP^{q+1}.$$

For $P > 0$,

$$\frac{dv}{dP} = k - (q+1)HP^q, \qquad \boxed{\frac{d^2v}{dP^2} = -q(q+1)HP^{q-1} < 0.}$$

The strict inequality follows from $q, H, P > 0$, even if $q < 1$. Hence there is at most one maximizer.

(b) Optimal debt and interior feasibility. Setting the first derivative to zero gives

$$(P^*)^q = \frac{k}{(q+1)H}.$$

Substituting for H,

$$\boxed{P^* = \frac{V}{\epsilon}\left[\frac{\kappa\rho/r}{(q+1)[\kappa\rho/r + (1-\alpha)\epsilon]}\right]^{1/q}.}$$

The bracket is strictly between 0 and 1 because its denominator exceeds its positive numerator. Therefore $0 < P^* < V/\epsilon$, so the implied barrier satisfies $\epsilon P^* < V$. Strict concavity proves that this stationary point is the unique global maximizer on the stated domain.

A useful value check follows from the first-order condition: $H(P^*)^{q+1} = kP^*/(q+1)$. Thus

$$\boxed{v(V; \epsilon P^*) = V + \frac{q}{q+1}kP^*.}$$

(c) Numerical evaluation and interpretation. Use $\epsilon = 0.8441527062$, $q = 1.8507810594$, $k = 0.30$, and $V = 150$:

$$\boxed{P^* = 67.1293, \qquad V_B^*(P^*) = 56.6674, \qquad v(V; V_B^*(P^*)) = 163.0745.}$$

These quantities obey the barrier feasibility condition. The same recovery coefficient as in Question 5 is $c = 0.6331145$, since $V_B^*/P^* = \epsilon$.

The smooth-pasting derivative is S_V at a fixed barrier and fixed debt level, evaluated at the default state. The second-stage derivative is of the total firm value with respect to the debt level, allowing the adopted barrier to change through $V_B^*(P) = \epsilon P$. The two derivatives address different decisions.

Solution to Question 8

(a) Survival and default probabilities. Survival to year-end j requires survival through j years:

$$Q_j = (1-p)^j.$$

Default at $t_j = j - 0.5$ requires survival through the first $j-1$ years and then default in year j:

$$d_j = (1-p)^{j-1}p.$$

Therefore

j	Q_j	d_j
1	0.975000000	0.025000000
2	0.950625000	0.024375000
3	0.926859375	0.023765625
4	0.903687891	0.023171484

The consistency check is $\sum_{j=1}^{4} d_j + Q_4 = 1$.

(b) Present values per unit notional. The regular premium paid at j has present value $sQ_je^{-0.04j}$. The default accrual at t_j has present value $0.5sd_je^{-0.04t_j}$, and the protection payment has present value $0.65d_je^{-0.04t_j}$.

Year	Regular premium coefficient	Accrual coefficient	Protection PV
1	0.93676970	0.01225248	0.01592823
2	0.87753748	0.01147776	0.01492108
3	0.82205052	0.01075201	0.01397762
4	0.77007202	0.01007216	0.01309381
Total	3.40642972	0.04455441	0.05792074

For example, the year-3 default probability is $0.975^2(0.025) = 0.023765625$, so its accrued-premium coefficient is $0.5(0.023765625)e^{-0.04(2.5)} = 0.01075201$. The first two columns must be multiplied by s.

(c) Break-even spread. Equate the buyer's regular and accrued premiums to the seller's protection:

$$s(3.40642972 + 0.04455441) = 0.05792074.$$

Using unrounded sums,

$$\boxed{s = 0.0167838315 = 1.678383\% = 167.8383 \text{ basis points.}}$$

The full annual premium on 10 million notional, conditional on survival to that payment date, is $10{,}000{,}000s = \boxed{167{,}838.31}$. On mid-year default, the accrued premium is half this amount.

(d) Effect of omitting accrual. Omitting accrual removes a positive term from the premium-leg coefficient. For an unchanged protection leg, the inferred spread is therefore too high.

Solution to Question 9

(a) Survival and cumulative default. For a deterministic intensity,

$$Q(T) = \exp\left(-\int_0^T \lambda(u)\,du\right).$$

The integrated intensities are

$$\int_0^2 \lambda(u)\,du = 0.04, \qquad \int_0^5 \lambda(u)\,du = 2(0.02) + 3(0.05) = 0.19.$$

Hence

T	Survival $Q(T)$	Cumulative default $1 - Q(T)$
2	$e^{-0.04} = 0.960789$	0.039211
5	$e^{-0.19} = 0.826959$	0.173041

(b) Bond prices, yields and spreads. With $1 - R = 0.60$,

$$B(0,2) = e^{-0.03(2)-0.60(0.04)} = e^{-0.084} = 0.919431,$$

$$B(0,5) = e^{-0.03(5)-0.60(0.19)} = e^{-0.264} = 0.767974.$$

For unit face value, $y(T) = -\log B(0,T)/T$, so

T	Price	Yield	Spread above r
2	0.919431	4.20%	1.20% = 120 bp
5	0.767974	5.28%	2.28% = 228 bp

The spread is the loss fraction $1-R$ times the average intensity over the maturity, not necessarily the intensity applying in the final year.

(c) Calibration. Let λ_1 apply over $(0,2]$ and λ_2 over $(2,5]$. Then

$$2[y(2) - r] = (1-R)2\lambda_1, \qquad 5[y(5) - r] = (1-R)(2\lambda_1 + 3\lambda_2).$$

Thus

$$\boxed{\lambda_1 = \frac{y(2) - r}{1 - R} = 0.02,}$$

$$\boxed{\lambda_2 = \frac{5[y(5) - r] - 2[y(2) - r]}{3(1 - R)} = 0.05.}$$

This calibration assumes the recovery fraction is known; prices alone do not identify recovery and intensity separately in this formula.

(d) Hazard sign and recovery. Because Q is decreasing, the default density is $f_\tau(T) = -Q'(T)$. Where the derivative exists,

$$\boxed{\lambda(T) = \frac{f_\tau(T)}{Q(T)} = -\frac{Q'(T)}{Q(T)}.}$$

For a fixed specified intensity, recovery does not enter the survival or default probabilities. It affects the loss on default, hence bond valuation. The recovery convention here is a fraction of pre-default market value; it differs from recovery of face value in Questions 2 and 8.

Source alignment and checks

This is an original practice set based on the supplied course materials. It reproduces their question types and core models, rather than their numerical data.

Questions	Supporting supplied materials
1	`midterm2026.pdf`, Q1; `lec1.pdf`, rating-transition estimation; `Assignment01-sol.pdf`, Q1.
2	`midterm2026.pdf`, Q2; `lec1.pdf`, implied default probabilities; `Assignment01-sol.pdf`, Q2.
3	`midterm2026.pdf`, Q3; `lec2.pdf`; `lec2_makeup.pdf`; `Assignment01-sol.pdf`, Q3.
4	`midterm2026.pdf`, Q4; `lec3.pdf`; `lec3_makeup.pdf`; `lec4.pdf`; `Assignment02-sol.pdf`, Q1.
5	`midterm2026.pdf`, Q5; `lec4.pdf`; `Assignment02-sol.pdf`, Q2.
6–7	`lec2.pdf`, Merton valuation and equity-volatility calibration; `lec4.pdf`, second-stage capital-structure optimization.
8–9	`lec5.pdf`, CDS valuation and Duffie–Singleton pricing (also reproduced in the latter part of `lec4.pdf`).

Conventions clarified in this set.

1. The bond-price exercise states the linear expected-loss approximation explicitly. Exact conditional probabilities require survival factors.

2. The annual probabilities in the CDS question are exact discrete conditional probabilities, rather than continuous-time hazards.

3. The lower-barrier Laplace transform uses the downward-passage sign.

4. The smooth-pasting bound includes an explicit proof of positivity. The displayed upper bound is not used to infer the distinct recovery restriction $c \leq 1$.

5. The hazard rate is $-Q'/Q$ when Q denotes survival. This sign ensures a nonnegative intensity for decreasing survival.

Numerical consistency checks. Generator rows sum to zero, transition rows sum to one, and default is absorbing. In Question 3, debt plus equity equals asset value plus tax benefit minus bankruptcy cost. In Question 4, equity is zero at the barrier. In Question 8, default probabilities plus terminal survival sum to one. Intermediate calculations retain full precision; printed values are rounded.